

Characteristics & Evolution of Computers

Why the machine is so capable, and how it developed over time

BSc Computer Science (First Year) · Detailed notes with diagrams · 2-mark & 14-mark questions

Learning objectives. After studying these notes you should be able to:

- List and explain the main characteristics of a computer.
- Explain GIGO and the real limitations of computers.
- Describe the development of computers before the electronic age.
- Describe the five generations and the technology defining each.
- State Moore's Law and the trend toward smaller, faster, cheaper machines.

1. Characteristics of a computer

These are the strengths that make computers so useful. Learn the list and a one-line explanation of each.

Characteristic	Meaning
Speed	Performs millions to billions of operations per second. Speed is measured in ms (10^{-3} s), μ s (10^{-6}), ns (10^{-9}) and ps (10^{-12}).
Accuracy	Gives exact, error-free results every time. Mistakes come from wrong input or logic, not the machine.
Diligence	Never tires, slows or gets bored; it repeats a task millions of times with the same speed and accuracy.
Versatility	One machine can do many different jobs simply by changing the program it runs.
Storage	Stores huge amounts of data and recalls any of it in a fraction of a second.
Automation	Once given a program, it carries out the whole task on its own without step-by-step human help.
Reliability	Produces consistent, dependable results over long periods.

2. Accuracy and GIGO

GIGO — Garbage In, Garbage Out. A computer's output is only as good as its input and instructions. If wrong data or wrong logic is supplied, the computer will process it flawlessly and still produce a wrong result.

This is a key idea: accuracy is a property of the **machine**, not of the data. The computer never makes an arithmetic error, so when results are wrong the fault lies with the human-supplied input or program.

3. Limitations of computers

- **No common sense** — it does exactly what it is told, even when that is obviously wrong.
- **No goals of its own** — a human must define every task; it cannot decide what to do.
- **No feelings or judgement** — it has no emotions, ethics, or awareness of consequences.
- **GIGO** — wrong input or logic gives wrong output.
- **Dependent** — without electricity and a program it does nothing at all.

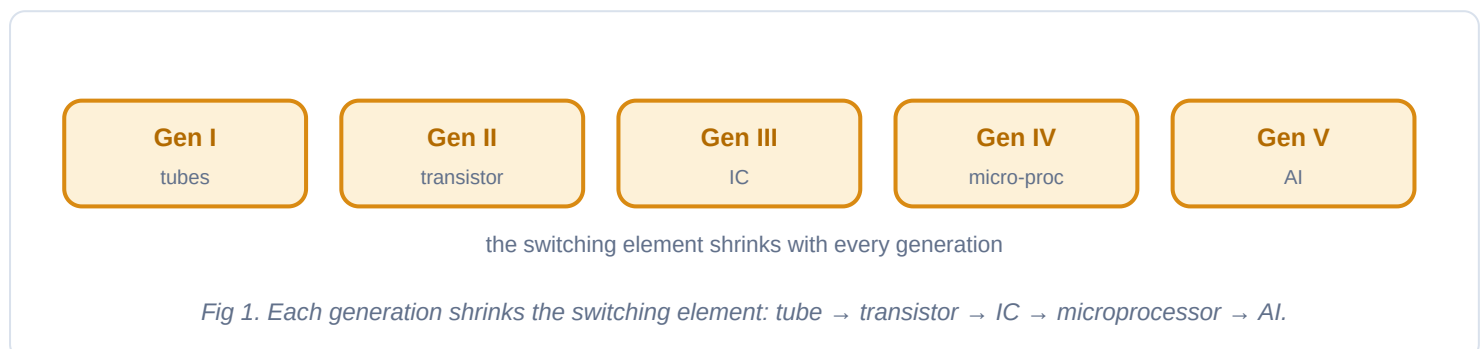
4. Before the electronic age

The idea of a calculating machine is ancient; only the electronics are new.

Year	Device	Contribution
c.2400 BC	Abacus	First manual counting/calculating tool.
1642	Pascaline (Pascal)	Mechanical adding machine using gears.
1801	Jacquard Loom	Punched cards controlled a machine — early programmability.
1837	Analytical Engine (Babbage)	Design of a general-purpose mechanical computer; Babbage is the 'father of the computer'.
1843	Ada Lovelace	Wrote the first algorithm — the first programmer.
1890	Hollerith Tabulator	Punched-card machine for the US census; led to IBM.

5. The five generations of computers

Modern computing is described in five **generations**, each defined by the technology used to switch a bit on and off.



Gen	Period	Technology	Examples	Features
I	1940–56	Vacuum tubes	ENIAC, UNIVAC I	Room-sized, very hot, unreliable, machine language.
II	1956–63	Transistors	IBM 1401, 7090	Smaller, faster, cooler, more reliable; FORTRAN, COBOL.
III	1964–71	Integrated circuits	IBM System/360	Cheaper; operating systems; keyboards & monitors.
IV	1971–present	Microprocessors (VLSI)	Intel 4004, IBM PC, Apple II	Whole CPU on one chip; personal computers; GUIs; internet.
V	present–future	AI, parallel, quantum	Smartphones, GPUs, AI chips	Massively parallel; machine learning; natural language.

6. Moore's Law and the shrinking machine

Moore's Law. Observed by Gordon Moore in 1965: the number of transistors that can be placed on a chip roughly **doubles about every two years**. This drives exponential growth in computing power.

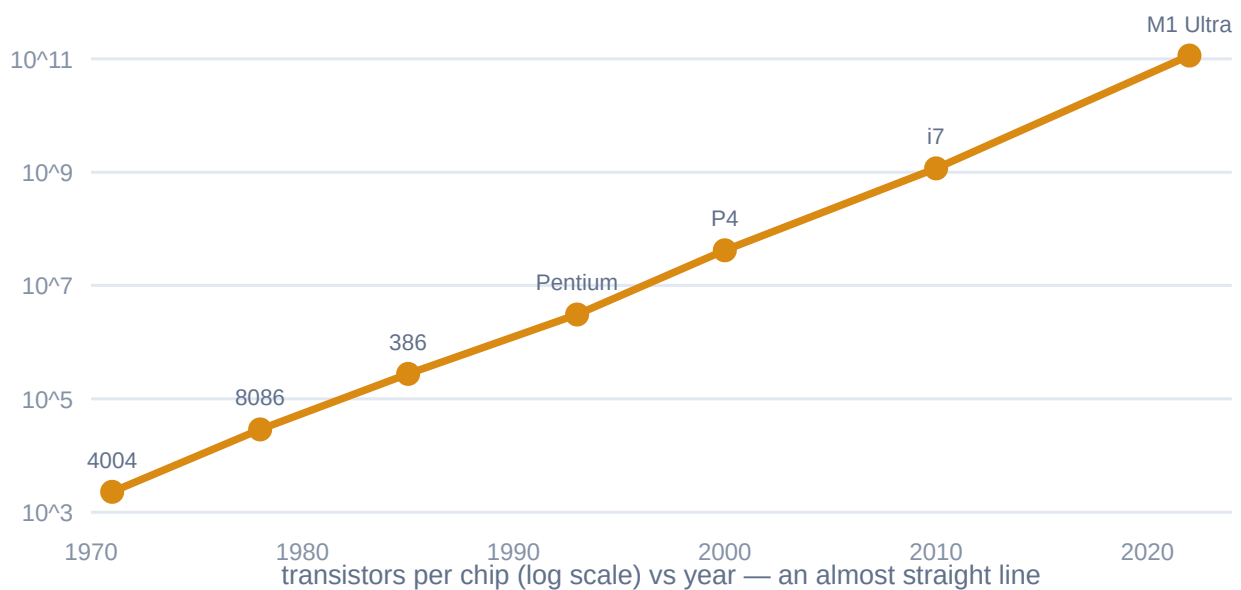


Fig 2. Transistor counts have grown from ~2,300 (1971) to over 100 billion — nearly a straight line on a log scale.

The overall trend of the generations is that computers became **smaller, faster, cheaper, more reliable and less power-hungry** — the same power that once filled a room now fits on a fingertip.

Chapter summary. Seven strengths (speed, accuracy, diligence, versatility, storage, automation, reliability); one big limitation (GIGO / no common sense); a mechanical past; five generations; and Moore's exponential curve.

2-Mark Questions (Short Answer)

Answer in 2–4 lines.

Q1. List any four characteristics of a computer.

2 marks

Speed, Accuracy, Diligence, Versatility (also Storage, Automation, Reliability).

Q2. What is meant by 'diligence' of a computer?

2 marks

Diligence means a computer can perform the same task repeatedly, for any length of time, without getting tired, bored or losing speed or accuracy.

Q3. What is GIGO?

2 marks

GIGO stands for Garbage In, Garbage Out — if wrong data or instructions are given, the computer will produce a wrong result even though it processed them perfectly.

Q4. State two limitations of a computer.

2 marks

It has no common sense and no ideas of its own — it only does what it is instructed. (Also: no feelings; depends on power and a program.)

Q5. Name the technology used in each of the first three generations.

2 marks

First generation: vacuum tubes; Second generation: transistors; Third generation: integrated circuits (ICs).

Q6. What defines the fourth generation of computers?

2 marks

The microprocessor — a complete CPU on a single chip (VLSI technology) — which led to the personal computer.

Q7. State Moore's Law.

2 marks

Moore's Law states that the number of transistors on a chip roughly doubles about every two years, giving exponential growth in computing power.

Q8. Who is called the 'father of the computer' and why?

2 marks

Charles Babbage, because he designed the Analytical Engine — the first design for a general-purpose mechanical computer.

Q9. Give one example each of a first- and a fourth-generation computer.

2 marks

First generation: ENIAC (or UNIVAC I). Fourth generation: IBM PC (or Apple II / Intel 4004-based machines).

Q10. Why is accuracy said to be a property of the machine and not the data?

2 marks

The computer itself never makes calculation errors; any wrong output is due to incorrect input or logic supplied by the user (GIGO).

14-Mark Questions (Long Answer)

Include definitions, well-structured points, a table or diagram where useful, and a conclusion.

Q1. Explain in detail the main characteristics of a computer.

14 marks

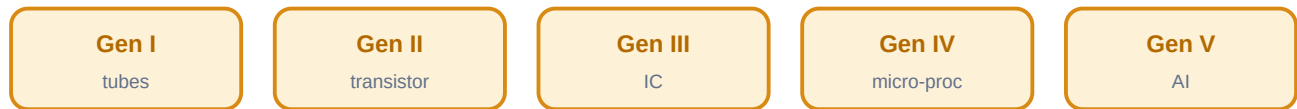
Introduction. A computer is powerful because of a set of specific strengths, called its characteristics. Together they explain why computers are trusted with critical work.

- **Speed.** A computer performs millions to billions of operations per second. Its speed is measured in very small units — milliseconds, microseconds, nanoseconds and even picoseconds. Work that would take a person years is finished in moments.
- **Accuracy.** Results are exact every time. The machine does not make arithmetic mistakes; errors come only from wrong input or logic (GIGO).
- **Diligence.** Unlike humans, a computer never tires, slows down or becomes bored. The millionth calculation is as fast and correct as the first.
- **Versatility.** The same machine can do completely different jobs — accounts, games, video editing, browsing — just by changing its program.
- **Storage.** It can store enormous amounts of data of every kind and retrieve any part almost instantly.
- **Automation.** Once a program is loaded, the computer carries out the whole task automatically, without a human guiding each step.
- **Reliability.** A well-built computer gives consistent, dependable results over long periods, which is why it is used in banking, medicine and aviation.

Conclusion. Speed, accuracy, diligence, versatility, storage, automation and reliability together make the computer an outstanding general-purpose tool — provided it is given correct data and instructions.

Q2. Describe the five generations of computers, mentioning the technology, examples and features of each. 14 marks

Introduction. The history of modern computers is divided into five generations. Each generation is marked by the technology used to switch bits on and off, and each is smaller, faster and cheaper than the last.



the switching element shrinks with every generation

The switching element shrinks across the five generations.

Gen	Period	Technology	Examples	Key features
First	1940–1956	Vacuum tubes	ENIAC, UNIVAC I	Very large (room-sized), generated much heat, unreliable, used machine language and punched cards.
Second	1956–1963	Transistors	IBM 1401, IBM 7090	Smaller, faster, more reliable and cooler than tubes; assembly and high-level languages (FORTRAN, COBOL).
Third	1964–1971	Integrated circuits	IBM System/360, PDP-8	Many transistors on one chip; cheaper and more compact; operating systems; keyboards and monitors.
Fourth	1971–present	Microprocessors (VLSI)	Intel 4004, IBM PC, Apple II	A whole CPU on a single chip; the personal computer; graphical interfaces, networks and the internet.
Fifth	present–future	AI, parallel & quantum	Smartphones, GPUs, AI accelerators	Massively parallel and increasingly intelligent — machine learning, natural language, vision.

Conclusion. From room-sized vacuum-tube machines to intelligent devices in our pockets, each generation shrank the switch and multiplied the power, following the trend of smaller, faster and cheaper computing.

Q3. What are the limitations of a computer? Explain the concept of GIGO with an example. 14 marks

Introduction. Although computers are extremely capable, they are tools with no mind of their own. Understanding their limitations is as important as knowing their strengths.

Limitations.

- **No common sense.** A computer follows instructions exactly, even when the result is obviously wrong to a human.
- **No ideas or goals of its own.** It cannot decide what problem to solve; every task must be defined by a person.
- **No feelings or judgement.** It has no emotions or ethics and cannot weigh the consequences of its output.
- **Cannot correct faulty logic.** If the program is wrong, the computer will not notice — it has no idea of the user's true intention.
- **Dependent on power and instructions.** Without electricity and a program it does nothing.

GIGO — Garbage In, Garbage Out. This principle states that the quality of the output depends entirely on the quality of the input and instructions. The computer processes whatever it is given flawlessly, so wrong input produces wrong output.

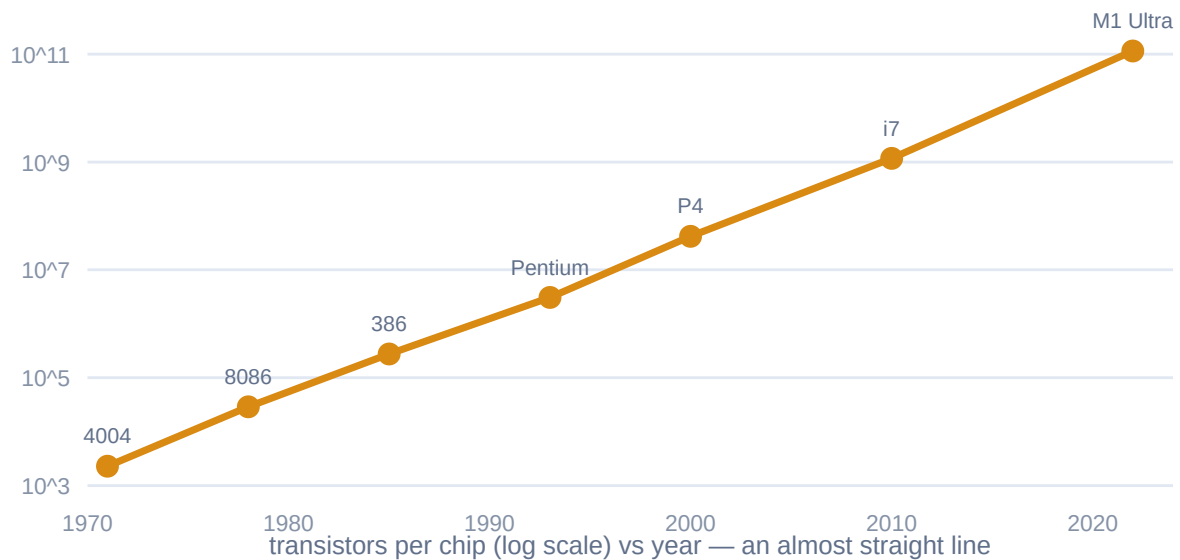
Example. If a shopkeeper enters a product price as 100 (with the letter O instead of the digit 0), the computer cannot treat it as the number 100. It will either reject it or miscalculate the bill — not because the machine made a mistake, but because the input was 'garbage'.

Conclusion. A computer is only as good as the data and logic given to it; GIGO reminds us that correct results require correct input.

Q4. Explain Moore's Law. How have computers changed over the generations in terms of size, speed and cost?

14 marks

Moore's Law. In 1965 Gordon Moore observed that the number of transistors that can be placed on an integrated circuit roughly doubles about every two years. Because more transistors means more computing power in the same space, this describes an **exponential** growth in capability that has held for decades.



Transistor count per chip has grown exponentially — a near-straight line on a log scale.

Changes across the generations.

- **Size.** Computers shrank from filling an entire room (first generation) to fitting in a pocket (fifth generation).
- **Speed.** Processing rose from thousands of operations per second to billions and trillions.
- **Cost.** The cost per calculation fell dramatically, so computing power that was once affordable only to governments is now in everyday devices.
- **Power and reliability.** Machines became far more energy-efficient and dependable as tubes gave way to transistors and then to dense chips.

Conclusion. Driven by Moore's Law, every generation delivered more power in less space, for less money and less energy — the defining trend of computing: smaller, faster and cheaper.